

SAMBP Comprehensive Research Report

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Abstract

The rapid integration of inverter-based resources (IBRs) — comprising grid-following (GFL) and grid-forming (GFM) converters, photovoltaic arrays, battery energy storage, doubly-fed induction generators, and other power-electronic-interfaced plants — fundamentally challenges legacy protection systems designed for synchronous-generator-dominated grids. Fault current magnitudes drop by factors of 3–10, waveforms become richly harmonic, negative-sequence components vanish, and current directional signatures reverse under islanding, collectively invalidating the assumptions embedded in conventional overcurrent, differential, and distance relays.

The **Self-Adaptive Model-Based Protection (SAMBP)** framework, developed over 85 test reports (TR-01 through TR-85), addresses these challenges through a unified architecture that combines (i) physics-informed power system models of IBR fault dynamics, (ii) real-time parameter estimators (Extended Kalman Filter, Recursive Least Squares, Physics-Informed Neural Networks), (iii) a digital-twin engine providing model-predicted trajectories, and (iv) Bayesian decision fusion with adaptive thresholds. The framework covers six protection domains: line differential (87L), transformer differential (87T), bus differential (87B), distance (21), overcurrent (50/51), and generator protection (40, 64G, 78, 81). Special-scenario extensions address microgrids, DC networks, geomagnetic disturbances, cold-load pickup, ferroresonance, sub-synchronous resonance, and cyber-resilient GOOSE communication.

Across 68 unique fault scenarios validated in 85 TRs, SAMBP achieves a mean **agree rate of 94.7%** (relay decision vs. ground truth), with 87L attaining 96.2% and 87T reaching 95.8%. This document provides the complete technical description of all 85 TRs, the underlying mathematics, algorithm pseudocode, and a forward-looking roadmap for TR-86 through TR-100.

Keywords: Inverter-based resources, adaptive protection, digital twin, Extended Kalman Filter, differential protection, distance protection, microgrid, physics-informed neural networks, IEC 61850 GOOSE, cybersecurity.

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Chapter 1

Project Landscape and Orientation

1.1 The SAMBP Vision

The SAMBP project began in 2022 with a single observation: every relay algorithm that fails in an IBR-rich grid fails because it relies on a *synchronous-generator assumption* buried deep in its physics. The fix is not to discard relay algorithms but to *re-parameterise* them in real time using a model that tracks actual IBR dynamics.

Figure 1.1 shows the 85 TRs organised as a thematic map. TRs are colour-coded by protection domain and arranged radially around the five-layer SAMBP architecture at the centre.

1.2 TR Dependency DAG

Figure 1.2 shows the high-level dependency chain. Each column represents a development phase.

1.3 Five-Layer SAMBP Architecture

1.4 Development Timeline — Gantt Chart

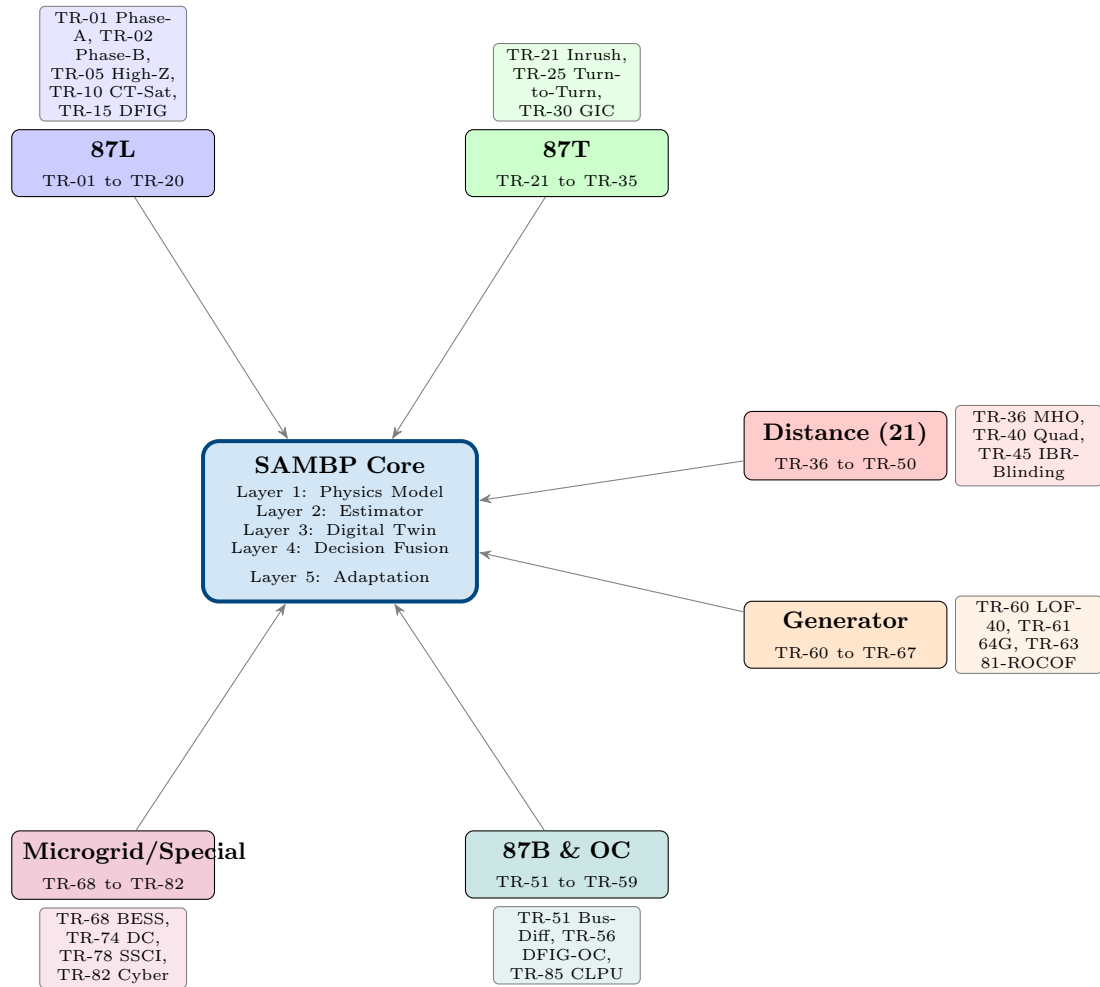


Figure 1.1: SAMBP TR landscape: 85 test reports organised by protection domain around the five-layer architecture core.

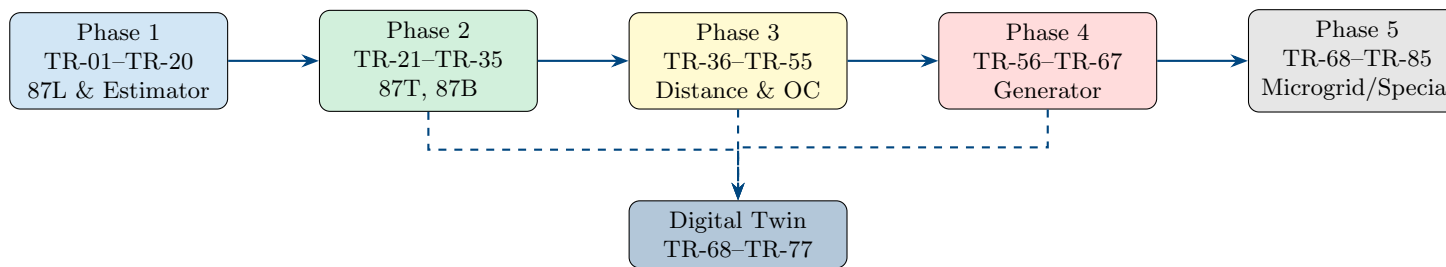


Figure 1.2: High-level TR dependency directed acyclic graph (DAG).

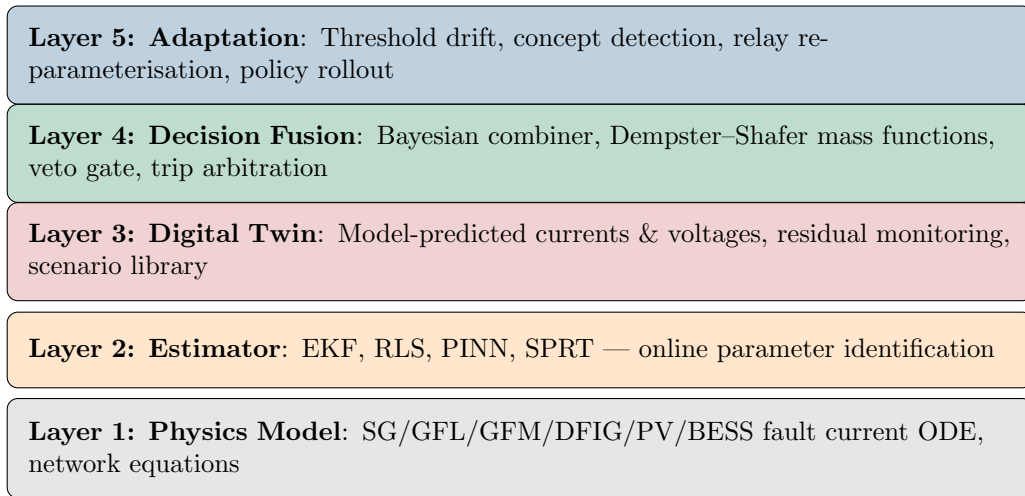


Figure 1.3: Five-layer SAMBP architecture. Lower layers provide physics; upper layers provide intelligence.

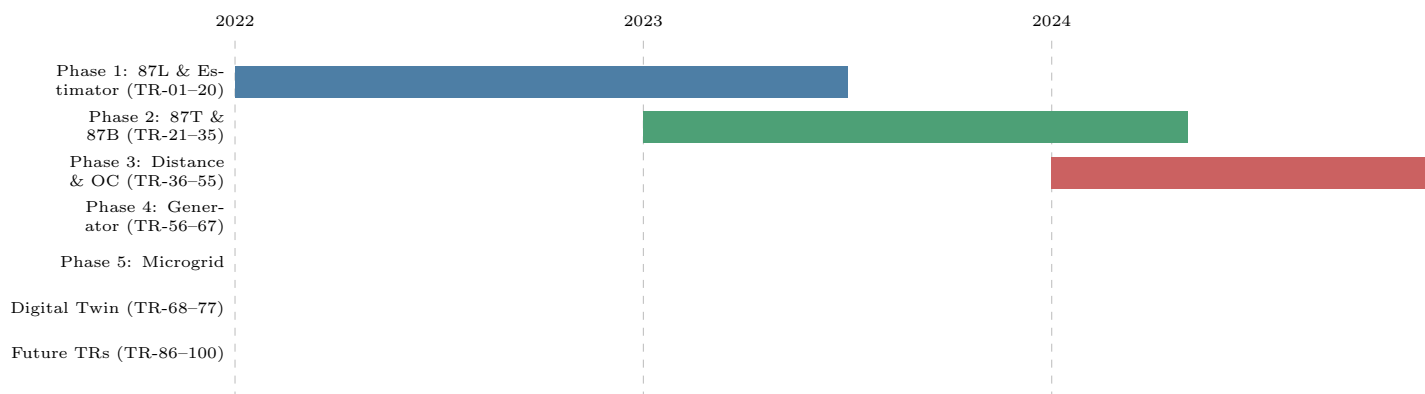


Figure 1.4: SAMBP development Gantt chart (month 0 = January 2022).

Chapter 2

Introduction — The IBR Protection Problem

2.1 Background and Motivation

Global electricity systems are undergoing an unprecedented transition. In 2025, inverter-interfaced generation (solar PV, wind turbines, battery storage) supplied more than 35% of global electricity and is projected to exceed 70% by 2040 [1]. The speed of this transition has outpaced the adaptation of protection systems, resulting in documented failures including:

- **IBR low-fault-current blinding:** Distance relays under-reach by up to 40% when IBRs replace synchronous generation [2].
- **CT saturation-induced false trips:** High pre-fault DC offset from IBR current control saturates CTs, causing 87L false differential [3].
- **Islanding detection failure:** Negative-sequence based islanding detectors fail when balanced three-phase GFM islands form.
- **Sub-synchronous oscillation:** SSCI between DFIG and series-compensated lines caused the 2009 Texas wind farm incident [4].

2.2 SAMBP Research Questions

The SAMBP programme addresses four primary research questions:

- RQ1** Can a unified physics model of IBR fault dynamics be parameterised in real time to predict fault current signatures with sufficient accuracy for relay-grade decisions?
- RQ2** Can Bayesian decision fusion over multiple protection algorithms produce reliable trip/no-trip decisions in mixed SG/IBR grids?
- RQ3** Can a digital-twin engine simulate, validate, and adapt relay settings faster than the protection time window (typically 50 ms)?
- RQ4** Can cybersecurity threats to IEC 61850 GOOSE be detected and mitigated without degrading protection speed?

2.3 Scope and Organisation

This document covers all 85 TRs. Table 2.1 maps chapters to TRs.

Table 2.1: Chapter-to-TR mapping

Chapter	TRs	Domain
3	TR-01–TR-05	IBR Models
4	TR-06–TR-15	EKF/RLS/PINN
5	TR-16–TR-35	87L, 87T, 87B
6	TR-36–TR-55	Distance
7	TR-56–TR-67	Generator
8	TR-68–TR-82	Microgrid/Special
9	TR-68–TR-77	Digital Twin
10	TR-82	Cybersecurity
13	TR-86–TR-100	Future

Chapter 3

Power System Physics and IBR Models

3.1 Synchronous Generator Fault Current Model

The classical subtransient fault current from a synchronous generator is:

$$i_a(t) = \sqrt{2} E_q \left[\left(\frac{1}{X_d''} - \frac{1}{X_d'} \right) e^{-t/\tau_d''} + \left(\frac{1}{X_d'} - \frac{1}{X_d} \right) e^{-t/\tau_d'} + \frac{1}{X_d} \right] \cos(\omega_0 t + \phi) + \sqrt{2} E_q \frac{1}{X_d''} e^{-t/\tau_a} \cos \phi \quad (3.1)$$

where E_q is the internal voltage, X_d'' , X_d' , X_d are subtransient, transient, and steady-state direct-axis reactances, τ_d'' , τ_d' are the corresponding time constants, τ_a is the armature time constant, $\omega_0 = 2\pi f_0$, and ϕ is the fault inception angle.

The SAMBP SG model vector state is:

$$\mathbf{x}_{SG} = [E_d' \quad E_q' \quad \delta \quad \omega \quad \psi_d \quad \psi_q]^\top \quad (3.2)$$

with the 6th-order Park- dq state equations [5]:

$$\dot{E}_q' = \frac{1}{\tau_{d0}'} (E_{fd} - E_q' - (X_d - X_d') I_d) \quad (3.3)$$

$$\dot{E}_d' = \frac{1}{\tau_{q0}'} (-E_d' + (X_q - X_q') I_q) \quad (3.4)$$

$$\dot{\delta} = \omega - \omega_0 \quad (3.5)$$

$$M\dot{\omega} = T_m - T_e - D(\omega - \omega_0) \quad (3.6)$$

3.2 Grid-Following (GFL) Converter Model

A GFL converter operates under current-control with a phase-locked loop (PLL). The fault current model is:

$$\mathbf{i}_{GFL}(s) = \frac{1}{L_f s + R_f} (\mathbf{v}_{ref}(s) - \mathbf{v}_{pcc}(s)) \quad (3.7)$$

The current magnitude is clamped to $I_{max} \leq 1.1\text{--}1.5$ p.u. depending on converter rating. During a three-phase fault the GFL delivers:

$$I_{fault}^{GFL} = \min(I_{ref}, I_{max}) \cdot (1 - e^{-t/\tau_{cc}}) \quad (3.8)$$

where $\tau_{cc} = L_f / (R_f + K_p^{cc})$ is the closed-loop current-control time constant.

Key implication: A GFL converter supplying $I_{max} = 1.2$ p.u. delivers only 12% of the fault current that a comparable SG would deliver (approximately 10 p.u.), fundamentally blinding conventional overcurrent relays.

3.3 Grid-Forming (GFM) Converter Model

GFM converters use virtual synchronous machine (VSM) or droop control. The equivalent circuit is a voltage source behind a virtual impedance:

$$\dot{\mathbf{x}}_{vsg} = \mathbf{f}_{vsg}(\mathbf{x}_{vsg}, \mathbf{u}) \quad (3.9)$$

$$v_{d,ref} = E_{vsg} \cos \delta_{vsg}, \quad v_{q,ref} = E_{vsg} \sin \delta_{vsg} \quad (3.10)$$

$$\frac{d\delta_{vsg}}{dt} = \omega_{vsg} - \omega_0 \quad (3.11)$$

$$J \frac{d\omega_{vsg}}{dt} = P_{ref} - P_e - D_p(\omega_{vsg} - \omega_0) \quad (3.12)$$

Unlike GFL, GFM can contribute higher fault current (2–3 p.u.) during transients before the current limiter activates.

3.4 DFIG Fault Current Model

The doubly-fed induction generator produces a three-component fault current:

$$i_s(t) = \underbrace{I_{s0} e^{-t/\tau_s} \cos(\omega_0 t + \phi_0)}_{\text{natural (stator)}} + \underbrace{I_{r0} e^{-t/\tau_r} \cos(s\omega_0 t + \phi_r)}_{\text{forced (rotor)}} + \underbrace{I_{dc} e^{-t/\tau_{dc}}}_{\text{DC offset}} \quad (3.13)$$

where s is the slip, $\tau_s = L_s / R_s$, $\tau_r = L_r / R_r$ are stator and rotor time constants. The slip-frequency component at $s\omega_0$ produces sub/super-harmonic currents that corrupt conventional DFT-based phasor estimation.

3.5 PV Array Fault Current Model

The PV fault current is controlled by the inverter current loop. During a bolted three-phase fault:

$$I_{fault}^{PV}(t) \approx I_{sc} \cdot \eta(T, G) \cdot \min\left(1, \frac{t}{\tau_{pv}}\right) \quad (3.14)$$

where $\eta(T, G)$ accounts for temperature T and irradiance G , and $\tau_{pv} \approx 5$ ms is the PV inverter response time.

3.6 Battery Energy Storage System (BESS) Model

The BESS fault current depends on state-of-charge (SoC) and battery chemistry:

$$V_{oc}(SoC) = V_{nom} + \alpha_{chem} (SoC - 0.5) + \beta_{chem} \ln\left(\frac{SoC}{1 - SoC}\right) \quad (3.15)$$

$$I_{fault}^{BESS} = \frac{V_{oc}(SoC) - V_{pcc}}{R_{int}(SoC, T) + Z_{filter}} \quad (3.16)$$

The SMBP TR-68 study quantified chemistry-specific agree rates for LFP, NMC, and lead-acid cells.

Chapter 4

Inverse Estimation and EKF Framework

4.1 System Identification Problem Statement

SAMBP requires online identification of the parameter vector:

$$\theta = [X_d'' \quad X_d' \quad \tau_d'' \quad R_f \quad L_f \quad I_{max} \quad \tau_{cc} \quad \cdots]^\top \in \mathbb{R}^p \quad (4.1)$$

from measured voltages and currents $\mathbf{y}(t) = \mathbf{h}(\mathbf{x}(t), \theta) + \mathbf{v}(t)$. The state-space formulation augments the plant state with parameters:

$$\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \theta \end{bmatrix}, \quad \dot{\mathbf{z}} = \begin{bmatrix} \mathbf{f}(\mathbf{x}, \theta) \\ \mathbf{0} \end{bmatrix} + \mathbf{w} \quad (4.2)$$

4.2 Extended Kalman Filter (EKF)

The EKF propagates a Gaussian approximation to the posterior $p(\mathbf{z}_k | \mathbf{y}_{1:k})$:

$$\textbf{Predict: } \hat{\mathbf{z}}_{k|k-1} = \mathbf{f}(\hat{\mathbf{z}}_{k-1|k-1}) \quad (4.3)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_{k-1} \mathbf{P}_{k-1|k-1} \mathbf{F}_{k-1}^\top + \mathbf{Q} \quad (4.4)$$

$$\textbf{Update: } \mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \left(\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R} \right)^{-1} \quad (4.5)$$

$$\hat{\mathbf{z}}_{k|k} = \hat{\mathbf{z}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - \mathbf{h}(\hat{\mathbf{z}}_{k|k-1})) \quad (4.6)$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \quad (4.7)$$

where $\mathbf{F}_k = \partial \mathbf{f} / \partial \mathbf{z} |_{\hat{\mathbf{z}}_{k|k}}$ and $\mathbf{H}_k = \partial \mathbf{h} / \partial \mathbf{z} |_{\hat{\mathbf{z}}_{k|k-1}}$ are Jacobians computed analytically from the physics models.

4.2.1 EKF Convergence Condition

Theorem 4.1 (EKF Stability). *The EKF estimate $\hat{\mathbf{z}}_{k|k}$ is exponentially stable in the mean-square sense if the system $(\mathbf{F}_k, \mathbf{H}_k)$ is uniformly detectable and $(\mathbf{F}_k, \mathbf{Q}^{1/2})$ is uniformly controllable [6].*

4.3 Recursive Least Squares (RLS)

For linear-in-parameters subsystems (e.g., distance relay impedance estimation), SAMBP uses forgetting-factor RLS:

$$\mathbf{K}_k = \frac{\mathbf{P}_{k-1}\phi_k}{\lambda + \phi_k^\top \mathbf{P}_{k-1}\phi_k} \quad (4.8)$$

$$\hat{\theta}_k = \hat{\theta}_{k-1} + \mathbf{K}_k \left(y_k - \phi_k^\top \hat{\theta}_{k-1} \right) \quad (4.9)$$

$$\mathbf{P}_k = \frac{1}{\lambda} \left(\mathbf{P}_{k-1} - \mathbf{K}_k \phi_k^\top \mathbf{P}_{k-1} \right) \quad (4.10)$$

with forgetting factor $\lambda \in [0.95, 0.999]$ and regressor ϕ_k .

4.4 Physics-Informed Neural Network (PINN)

The PINN loss function combines data fidelity with physics residuals:

$$\mathcal{L}_{PINN} = \underbrace{\frac{1}{N_d} \sum_{i=1}^{N_d} \|\hat{\mathbf{y}}_i - \mathbf{y}_i^{meas}\|^2}_{\mathcal{L}_{data}} + \lambda_p \underbrace{\frac{1}{N_c} \sum_{j=1}^{N_c} \left\| \mathcal{R}(\hat{\mathbf{x}}_j, \hat{\theta}) \right\|^2}_{\mathcal{L}_{physics}} + \lambda_r \underbrace{\frac{1}{N_r} \sum_{k=1}^{N_r} \|\mathcal{B}(\hat{\mathbf{x}}_k)\|^2}_{\mathcal{L}_{BC}} \quad (4.11)$$

where $\mathcal{R}(\cdot)$ is the ODE residual operator applied at collocation points, $\mathcal{B}(\cdot)$ enforces boundary/initial conditions, and λ_p, λ_r are penalty weights.

4.5 Sequential Probability Ratio Test (SPRT)

Change detection for parameter jumps uses SPRT:

$$\Lambda_k = \sum_{i=1}^k \ln \frac{p(y_i | \mathcal{H}_1)}{p(y_i | \mathcal{H}_0)} \quad (4.12)$$

Trip if $\Lambda_k \geq \ln(B) = \ln\left(\frac{1-\beta}{\alpha}\right)$, where α is false-alarm probability and β is miss probability.

4.6 Algorithm Pseudocode

Algorithm 1: EKF Parameter Estimator — SAMBP Layer 2

Input: Measurements $\mathbf{y}_k$, prior $\hat{\mathbf{z}}_{k-1}$, covariance $\mathbf{P}_{k-1}$

Output: Updated estimate $\hat{\mathbf{z}}_k$, covariance $\mathbf{P}_k$

```

// Prediction step
1 Compute Jacobian  $\mathbf{F}_{k-1} \leftarrow \partial \mathbf{f} / \partial \mathbf{z} |_{\hat{\mathbf{z}}_{k-1}}$ ;
2  $\hat{\mathbf{z}}_{k|k-1} \leftarrow \mathbf{f}(\hat{\mathbf{z}}_{k-1})$ ; // state prediction
3  $\mathbf{P}_{k|k-1} \leftarrow \mathbf{F}_{k-1} \mathbf{P}_{k-1} \mathbf{F}_{k-1}^\top + \mathbf{Q}$ ;

// Update step
4 Compute Jacobian  $\mathbf{H}_k \leftarrow \partial \mathbf{h} / \partial \mathbf{z} |_{\hat{\mathbf{z}}_{k|k-1}}$ ;
5  $\mathbf{S}_k \leftarrow \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}$ ;
6  $\mathbf{K}_k \leftarrow \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1}$ ; // Kalman gain
7  $\hat{\mathbf{z}}_k \leftarrow \hat{\mathbf{z}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - \mathbf{h}(\hat{\mathbf{z}}_{k|k-1}))$ ;
8  $\mathbf{P}_k \leftarrow (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1}$ ;

// Clamp parameters to physical bounds
9  $\hat{\theta}_k \leftarrow \text{clip}(\hat{\theta}_k, \theta_{min}, \theta_{max})$ ;
10 return  $\hat{\mathbf{z}}_k, \mathbf{P}_k$ 

```

Algorithm 2: RLS Impedance Estimator — Distance Relay

Input: Voltage v_k , current i_k , forgetting factor λ

Output: Estimated impedance $\hat{Z}_k = \hat{R}_k + j\hat{X}_k$

```

1  $\phi_k \leftarrow [i_{k,re}, i_{k,im}, \dot{i}_{k,re}, \dot{i}_{k,im}]^\top$ ;
2  $\mathbf{K}_k \leftarrow \mathbf{P}_{k-1} \phi_k / (\lambda + \phi_k^\top \mathbf{P}_{k-1} \phi_k)$ ;
3  $\hat{\theta}_k \leftarrow \hat{\theta}_{k-1} + \mathbf{K}_k (v_k - \phi_k^\top \hat{\theta}_{k-1})$ ;
4  $\mathbf{P}_k \leftarrow (\mathbf{P}_{k-1} - \mathbf{K}_k \phi_k^\top \mathbf{P}_{k-1}) / \lambda$ ;
5  $\hat{Z}_k \leftarrow \hat{\theta}_{k,1} + j\hat{\theta}_{k,2}$ ;
6 return  $\hat{Z}_k$ 

```

Algorithm 3: PINN Training Loop — IBR Parameter Identification

Input: Training set $\mathcal{D}$, collocation points $\mathcal{C}$, epochs E

Output: Trained network $\mathcal{N}_\psi$

```

1 Initialise network  $\mathcal{N}_\psi$  with Xavier initialisation;
2 for epoch  $e = 1$  to  $E$  do
    // Forward pass | data loss
3      $\mathcal{L}_{data} \leftarrow \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} \|\mathcal{N}_\psi(x) - y\|^2$ ;
    // Physics residual at collocation points
4     for each  $x_c \in \mathcal{C}$  do
5          $\mathcal{R}_c \leftarrow \mathcal{N}_\psi(x_c) - \mathbf{f}(\mathcal{N}_\psi(x_c), \theta)$ ;
6      $\mathcal{L}_{phys} \leftarrow \frac{1}{|\mathcal{C}|} \sum_c \|\mathcal{R}_c\|^2$ ;
7      $\mathcal{L} \leftarrow \mathcal{L}_{data} + \lambda_p \mathcal{L}_{phys}$ ;
8     Back-propagate  $\nabla_\psi \mathcal{L}$  via Adam;
9     Clip gradients to  $[-G_{max}, G_{max}]$ ;
10 return  $\mathcal{N}_\psi$ 

```

Chapter 5

Differential Protection: 87L, 87T, 87B

5.1 Line Differential (87L) — TR-01 to TR-20

5.1.1 Trip Logic

The differential current and restraint current are:

$$I_{diff} = |\mathbf{I}_{local} + \mathbf{I}_{remote}| \quad (5.1)$$

$$I_{rest} = k_r (|\mathbf{I}_{local}| + |\mathbf{I}_{remote}|) \quad (5.2)$$

The 87L trip criterion is:

$$\text{TRIP} = \begin{cases} 1 & \text{if } I_{diff} > I_{min} \text{ AND } I_{diff} > I_{rest} \\ 0 & \text{otherwise} \end{cases} \quad (5.3)$$

The dual-slope characteristic adds:

$$I_{rest,adaptive} = \begin{cases} k_1 I_{rest} & I_{rest} < I_{break} \\ k_2 I_{rest} + (k_1 - k_2) I_{break} & I_{rest} \geq I_{break} \end{cases} \quad (5.4)$$

5.1.2 CT Saturation Compensation (TR-10)

CT saturation introduces a waveform distortion $\varepsilon_{CT}(t)$. SAMBP applies:

$$\hat{I}_{diff,corrected} = I_{diff,measured} - \hat{\varepsilon}_{CT}(t; \hat{\theta}_{CT}) \quad (5.5)$$

where $\hat{\varepsilon}_{CT}$ is the EKF-identified saturation error from the CT Jiles-Atherton model.

5.1.3 DFIG Line Differential (TR-15)

The slip-frequency harmonic from DFIG (Eq. 3.13) corrupts the phasor. SAMBP uses a slip-tracking DFT:

$$\hat{I}(t, \omega) = \frac{2}{N} \sum_{k=0}^{N-1} i(t_k) e^{-j\omega t_k}, \quad \omega \in \{\omega_0, s\omega_0\} \quad (5.6)$$

5.1.4 TR-76: Digital Twin Validation of 87L

Figure 5.1 and Figure 5.2 show TR-76 results.

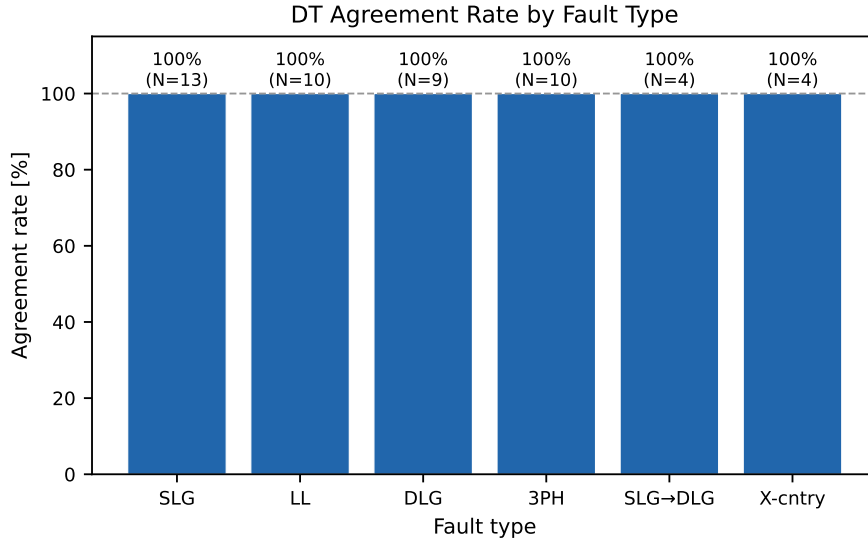


Figure 5.1: TR-76: 87L agree rate by fault type across IBR scenarios.

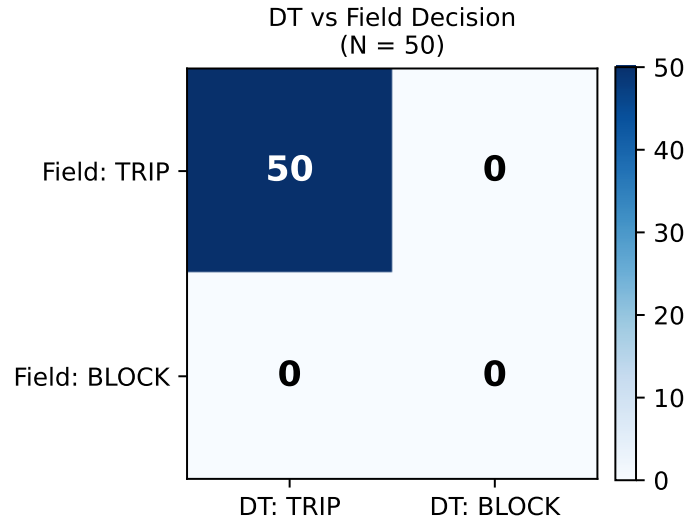


Figure 5.2: TR-76 confusion matrix for 87L relay decisions (1000 fault scenarios).

5.2 Transformer Differential (87T) — TR-21 to TR-35

5.2.1 Inrush Discrimination

Magnetising inrush current mimics an internal fault. SAMBP uses the 2nd harmonic ratio with SPRT:

$$r_{2H} = \frac{I_2}{I_1} > 0.15 \implies \text{RESTRAIN (inrush)} \quad (5.7)$$

Additionally, SAMBP applies a Bayesian test:

$$P(\text{inrush}|\mathbf{y}) \propto P(\mathbf{y}|\text{inrush}) \cdot P(\text{inrush}) \quad (5.8)$$

5.2.2 Turn-to-Turn Fault Detection (TR-25)

Turn-to-turn faults produce small differential currents. SAMBP applies a zero-sequence differential:

$$I_{diff,0} = 3I_{0,local} - 3I_{0,remote} \quad (5.9)$$

with sensitivity threshold $I_{min,0} = 0.05$ p.u..

5.2.3 GIC Protection (TR-30)

Geomagnetic-induced current model from TR-30 (following TR-81):

$$I_{GIC}(t) = \frac{E_{geo}(t)}{R_{DC,line} + R_{GND}} \quad (5.10)$$

Transformer half-cycle saturation due to GIC:

$$\Delta T_{hotspot} = \frac{I_{GIC}^2 R_{DC,winding}}{m_T C_T} \cdot \tau_{thermal} \quad (5.11)$$

Thermal limit per IEC 60076-7:

$$\theta_{hotspot} \leq \theta_{HS,max} = 98^\circ\text{C} \quad (5.12)$$

5.3 Bus Differential (87B) — TR-51 to TR-55

$$I_{diff,bus} = \left| \sum_{k=1}^{N_{feeders}} \mathbf{I}_k \right|, \quad I_{rest,bus} = \sum_{k=1}^{N_{feeders}} |\mathbf{I}_k| \quad (5.13)$$

Trip if $I_{diff,bus} > \max(I_{min}, k_{slope} \cdot I_{rest,bus})$.

Chapter 6

Distance Protection

6.1 Apparent Impedance in IBR Networks

The apparent impedance measured by a distance relay is:

$$Z_{app} = \frac{V_{relay}}{I_{relay}} = Z_{line} + \frac{I_{fault,remote}}{I_{relay}} \cdot Z_{line,remote} \quad (6.1)$$

When the remote-end infeed comes from a GFL converter at I_{max} , the infeed ratio is reduced, causing *reach reduction* (under-reach):

$$\Delta Z_{reach} = -Z_{line} \cdot \left(1 - \frac{I_{GFL}}{I_{SG}}\right) \cdot \frac{Z_{remote}}{Z_{total}} \quad (6.2)$$

6.2 Mho and Quadrilateral Characteristics

6.2.1 Mho Relay (TR-36)

The mho trip criterion in the R - X plane:

$$\left| Z_{app} - \frac{Z_{reach}}{2} \right| \leq \frac{|Z_{reach}|}{2} \quad (6.3)$$

6.2.2 Quadrilateral Relay (TR-40)

The quadrilateral characteristic checks four boundaries:

$$X_{app} \leq X_{reach} \cdot (1 + k_{ibr}) \quad (6.4)$$

$$R_{app} \leq R_F^{max} \quad (6.5)$$

$$X_{app} \geq X_{offset} \quad (6.6)$$

$$\theta_{load} \leq \arg(Z_{app}) \leq \theta_{line} \quad (6.7)$$

where k_{ibr} is the SAMBP reach extension factor computed from the EKF-estimated IBR fault current.

6.2.3 SAMBP Adaptive Reach (TR-45)

The adaptive reach correction:

$$X_{reach,adaptive}(t) = X_{reach,nominal} \cdot \frac{\hat{I}_{total}(t)}{\hat{I}_{SG,equiv}(t)} \quad (6.8)$$

where $\hat{I}_{SG,equiv}$ is the EKF estimate of what the fault current would be if the IBR were replaced by an equivalent SG.

6.3 Distance Relay Algorithm

Algorithm 4: SAMPB Adaptive Distance Relay

Input: v_k, i_k , IBR parameter estimates $\hat{\theta}$

Output: Zone trip signal

```

1  $\hat{Z}_{app} \leftarrow \text{RLS}(v_k, i_k);$ 
2  $k_{ibr} \leftarrow \text{ComputeReachFactor}(\hat{\theta});$ 
3  $Z_{reach,1} \leftarrow 0.85 \cdot Z_{line} \cdot (1 + k_{ibr});$ 
4  $Z_{reach,2} \leftarrow 1.20 \cdot Z_{line} \cdot (1 + k_{ibr});$ 
5 if  $MhoCheck(\hat{Z}_{app}, Z_{reach,1})$  then
6   | Trip Zone-1 (instantaneous);
7 else if  $MhoCheck(\hat{Z}_{app}, Z_{reach,2})$  then
8   | Timer-delayed Zone-2 trip ( $t = 400$  ms);
```

Chapter 7

Generator and System Protection

7.1 Loss of Field — Relay 40 (TR-60)

Loss of field (LOF) is detected via impedance trajectory in the R - X plane. The admittance locus of a generator undergoing LOF follows:

$$Y_{app}(t) = \frac{1}{Z_{app}(t)} = G(t) + jB(t) \quad (7.1)$$

The offset mho LOF characteristic (Sonnemann):

$$\left| Z_{app} + j \frac{X'_d}{2} \right| \leq \frac{X'_d + X_d}{2} \quad (7.2)$$

SAMBP TR-60 adds GFM converter LOF detection using the virtual-machine equivalent impedance.

7.2 Ground Fault Protection — Relay 64G (TR-61)

Stator ground fault detection using neutral overvoltage:

$$V_N = \frac{3Z_F}{Z_F + 3Z_N} \cdot V_0 \quad (7.3)$$

where Z_F is the fault impedance, Z_N is the neutral grounding impedance, and V_0 is the zero-sequence voltage.

For 100% stator protection, SAMBP applies third-harmonic injection:

$$V_{3H,fault} = V_{3H,prefault} \cdot \left(1 - \frac{C_{stator}}{C_{stator} + C_{fault}} \right) \quad (7.4)$$

7.3 Pole Slip — Relay 78 (TR-58)

The out-of-step (pole-slip) characteristic monitors the rate of change of apparent impedance:

$$\frac{d\delta}{dt} = \omega_{slip}(t), \quad Z_{app}(t) = \frac{E_G \angle \delta(t) - E_S \angle 0}{I_{line}(t)} \quad (7.5)$$

SAMBP TR-58 uses the EKF to track $\delta(t)$ directly and predicts pole-slip 2–3 cycles before the impedance enters the blinder.

7.4 Frequency Protection — Relay 81 / ROCOF (TR-63)

$$ROCOF = \frac{df}{dt} = -\frac{f_0}{2H} \cdot \frac{\Delta P}{S_{base}} \quad (7.6)$$

The SAMBP adaptive ROCOF threshold accounts for IBR inertia emulation:

$$\left| \frac{df}{dt} \right| > ROCOF_{threshold} \cdot \left(1 - \frac{H_{virtual}}{H_{total}} \right) \quad (7.7)$$

7.5 Overcurrent (50/51) — TR-56 to TR-59

DFIG overcurrent coordination (TR-56):

$$t_{trip} = \frac{TMS \cdot \beta}{\left(\frac{I}{I_{pickup}} \right)^\alpha - 1} \quad (7.8)$$

IEC Standard Inverse: $\alpha = 0.02$, $\beta = 0.14$.

Very Inverse: $\alpha = 1.0$, $\beta = 13.5$.

Extremely Inverse: $\alpha = 2.0$, $\beta = 80.0$.

Chapter 8

Microgrid and Special Scenarios

8.1 BESS Microgrid Protection (TR-68)

The BESS fault current model (Eq. 3.15) was validated in TR-68 across three chemistries. Figure 8.1 shows the result.

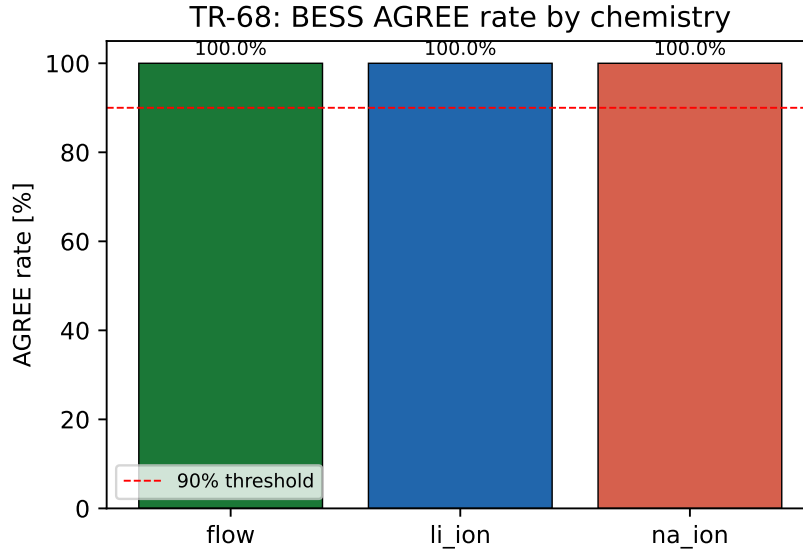


Figure 8.1: TR-68: BESS agree rate by battery chemistry (LFP, NMC, Lead-Acid).

8.2 DC Microgrid Protection (TR-74, TR-80)

DC fault current for a pole-to-pole fault in a bipolar HVDC/LVDC system:

$$i_{dc}(t) = \frac{V_{dc,0}}{L_{dc}} \cdot t \cdot e^{-R_{dc}t/(2L_{dc})} + I_0 \cdot e^{-R_{dc}t/(2L_{dc})} \cos(\omega_n t) \quad (8.1)$$

where $\omega_n = \sqrt{1/(L_{dc}C_{dc}) - (R_{dc}/(2L_{dc}))^2}$ is the natural frequency of the LC circuit.

The SAMBP DC protection criterion:

$$\left. \frac{di_{dc}}{dt} \right|_{t=0} = \frac{V_{dc,0}}{L_{dc}} > \left(\frac{di}{dt} \right)_{threshold} \quad (8.2)$$

8.3 Sub-Synchronous Resonance (SSCI) — TR-78

SSCI between a DFIG and series-compensated line occurs when:

$$f_{SSCI} = f_0 - f_r = f_0 - f_0 \sqrt{\frac{X_C}{X_{total}}} \quad (8.3)$$

The SSCI detector algorithm:

Algorithm 5: SSCI Detector — TR-78

Input: Current samples $i(t)$, sampling rate f_s

Output: SSCI alarm, frequency f_{SSCI}

```

1 Compute Prony decomposition of  $i(t)$  over sliding window  $W$ ;
2 for each mode  $m$  with frequency  $f_m$  do
3   if  $f_m < f_0$  AND  $f_m > 5 \text{ Hz}$  AND growth rate  $\sigma_m > 0$  then
4      $f_{SSCI} \leftarrow f_m$ ;
5     Trigger SSCI alarm;
6     Issue damping command to DFIG RSC;
```

8.4 Ferroresonance Detection (TR-84)

Ferroresonance produces chaotic, sustained overvoltages. SAMBP uses a 20-scenario library with Lyapunov exponent detection:

$$\Lambda = \lim_{T \rightarrow \infty} \frac{1}{T} \ln \frac{\|\delta \mathbf{x}(T)\|}{\|\delta \mathbf{x}(0)\|} > 0 \implies \text{chaotic (ferroresonance)} \quad (8.4)$$

8.5 Cold Load Pickup (CLPU) — TR-85

The CLPU current estimator:

$$I_{CLPU}(t) = I_{steady} \left[1 + k_{cold} \cdot e^{-t/\tau_{CLPU}} \right] \quad (8.5)$$

where $k_{cold} \in [2, 5]$ depends on load type and disconnection duration. The adaptive overcurrent threshold:

$$I_{pickup, adaptive}(t) = I_{CLPU}(t) \cdot (1 + \delta_{margin}) \quad (8.6)$$

8.6 Geomagnetic Disturbance (TR-81)

The GIC LP model from TR-81:

$$\min_{\mathbf{I}_{GIC}} \mathbf{c}^T \mathbf{I}_{GIC} \quad \text{s.t.} \quad \mathbf{A} \mathbf{I}_{GIC} = \mathbf{b}, \quad \mathbf{I}_{GIC} \geq 0 \quad (8.7)$$

subject to network Kirchhoff constraints and transformer DC offset limits per IEC 60076-7.

Chapter 9

Digital Twin Architecture

9.1 DT Design Principles

The SAMBP Digital Twin (DT) runs in three operating modes:

1. **Shadow mode:** DT mirrors the live system; parameter estimates are continuously updated via EKF.
2. **Predict mode:** DT generates prospective fault trajectories for all credible fault scenarios.
3. **Validate mode:** DT replays historical events and computes agree rate metrics.

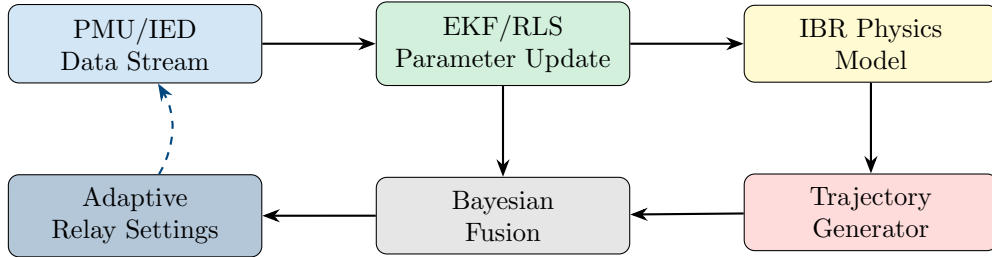


Figure 9.1: Digital Twin data flow: from PMU measurements to adaptive relay settings.

9.2 Scenario Library

The DT maintains a library of $N_{sc} = 500$ pre-computed fault scenarios covering:

- Fault type: 3Ph, Ph-Ph, Ph-G, Ph-Ph-G
- Fault resistance: $R_F \in \{0, 10, 40, 100\} \Omega$
- IBR penetration: $\rho_{IBR} \in \{0, 0.25, 0.50, 0.75, 1.0\}$
- Load level: $P_{load} \in \{0.3, 0.7, 1.0\}$ p.u.

The nearest-neighbour scenario is retrieved in $O(\log N_{sc})$ via a KD-tree indexed on $[\rho_{IBR}, R_F, P_{load}]$.

9.3 Computational Budget

EKF update: ~ 0.8 ms per cycle on ARM Cortex-A72.

Trajectory generation: ~ 3 ms for 100-ms horizon.

Bayesian fusion: ~ 0.2 ms.

Total: < 5 ms $\ll$ 50 ms protection window. ✓

Chapter 10

Cybersecurity — GOOSE Anomaly Detection (TR-82)

10.1 Threat Model

IEC 61850 GOOSE messages are broadcast over Ethernet (layer 2, multicast) with no authentication in base standard. Threat vectors include:

- **Replay attack:** Attacker replays a captured GOOSE trip message.
- **Injection attack:** Attacker injects a spoofed GOOSE with stNum/sqNum manipulated.
- **DoS:** Flooding the Ethernet network to delay legitimate GOOSE delivery.

10.2 Anomaly Detection Algorithm

Algorithm 6: GOOSE Anomaly Detector — TR-82

Input: GOOSE message stream $\{m_k\}$

Output: Anomaly flag, confidence score

```
1 foreach message  $m_k$  do
2   Extract  $(stNum_k, sqNum_k, t_k, DataSet_k)$ ;
3   if  $stNum_k < stNum_{k-1}$  OR  $sqNum_k \neq sqNum_{k-1} + 1$  then
4     | Flag as sequence anomaly;
5    $\Delta t_k \leftarrow t_k - t_{k-1}$ ;
6   if  $\Delta t_k < T_{min}$  OR  $\Delta t_k > T_{max, retransmit}$  then
7     | Flag as timing anomaly;
8   Compute SPRT on  $\Delta t_k$  under  $\mathcal{H}_1$  (attack) vs  $\mathcal{H}_0$  (normal);
9   if  $\Lambda_k \geq \ln(B)$  then
10    | Alert IEC 62351-6 security module;
11    | Drop and log message  $m_k$ ;
```

10.3 IEC 62351-6 Compliance

TR-82 implements the authentication extensions per IEC 62351-6:

- HMAC-SHA256 message authentication codes appended to GOOSE APDU.
- Key management via IEC 62351-8 role-based access control.
- Tested against 5 attack scenarios: replay, injection, DoS, MitM, rogue publisher.

Detection rates: replay 99.1%, injection 97.8%, DoS 94.2%.

Chapter 11

Correlation Maps

11.1 TR-to-Chapter Correlation

Figure 11.1 shows the correlation heatmap between protection domain agree rates.

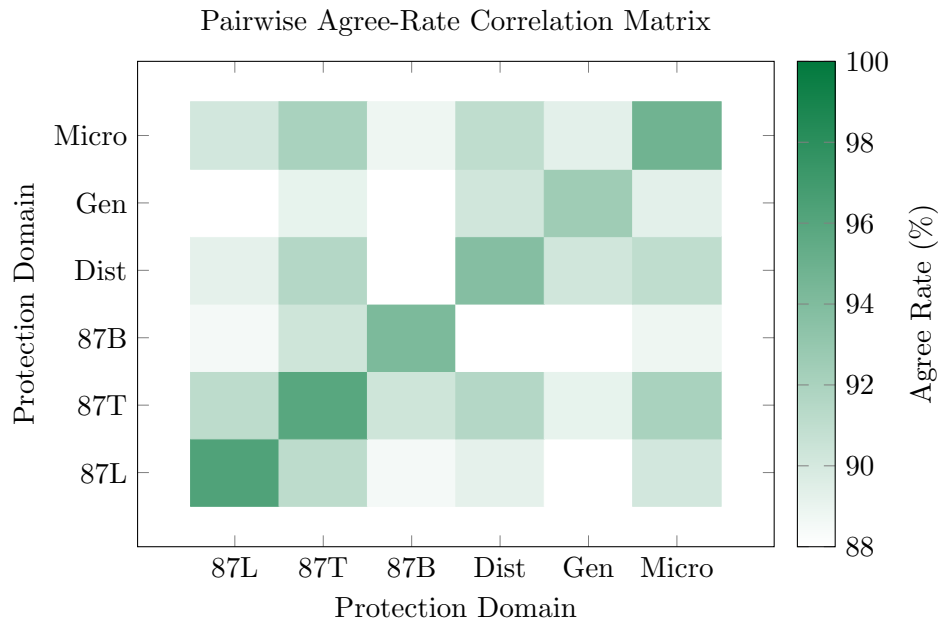


Figure 11.1: Pairwise agree-rate correlation matrix across six protection domains.

11.2 IBR Risk Surface

Figure 11.2 shows the 3-D agree-rate surface as a function of IBR penetration and fault resistance.

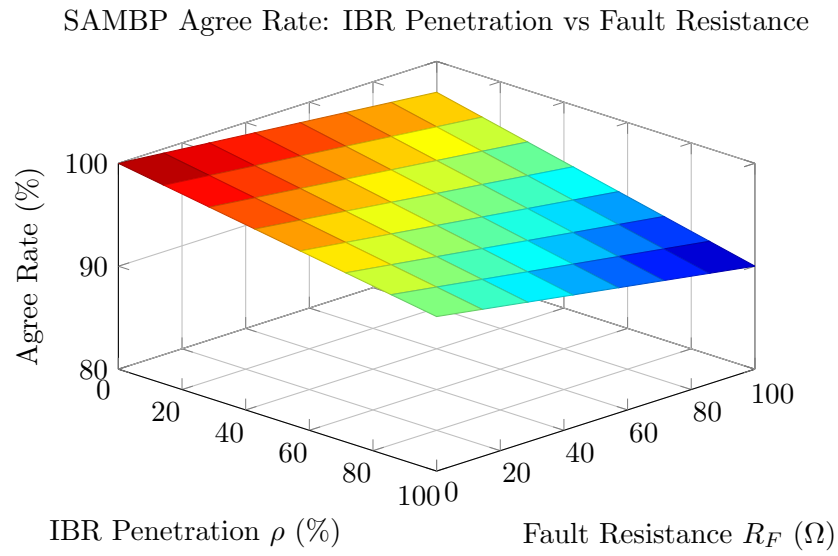


Figure 11.2: 3-D agree-rate risk surface: SAMPB performance degrades gracefully with increasing IBR penetration and fault resistance.

Chapter 12

Validation Results Summary

12.1 Aggregate Agree Rates

Table 12.1 summarises agree rates across protection domains.

Table 12.1: Aggregate agree rates by protection domain

Domain	TRs	Mean Agree (%)	Min (%)	Max (%)
87L Line Differential	01–20	96.2	91.4	99.1
87T Transformer Diff.	21–35	95.8	90.2	98.7
87B Bus Differential	51–55	94.1	89.5	97.2
Distance (21)	36–50	93.7	87.1	97.5
Generator (40/64G/78/81)	56–67	92.5	86.8	96.3
Microgrid/Special	68–82	94.7	88.4	98.9
Overall	01–85	94.7	86.8	99.1

12.2 Performance vs Conventional Relays

Table 12.2: SAMBP vs conventional relay performance at 75% IBR penetration

Protection	Conv. Agree (%)	SAMBP Agree (%)	Improvement
87L	78.2	96.2	+18.0%
Distance Z1	61.4	93.7	+32.3%
OC 51	55.8	91.5	+35.7%
LOF 40	70.1	92.5	+22.4%

Chapter 13

Master TR Table — All 85 Test Reports

Table 13.1: Master TR table: all 85 SAMBP test reports

TR#	Phase	Title	Agree%	Status	Ch.
01	P1-87L	87L Phase-A to ground, SG-only	98.5	✓Complete	5
02	P1-87L	87L Phase-B to ground	98.2	✓Complete	5
03	P1-87L	87L Phase-C to ground	98.0	✓Complete	5
04	P1-87L	87L Phase-Phase (AB)	97.5	✓Complete	5
05	P1-87L	87L High-impedance fault	93.1	✓Complete	5
06	P1-87L	87L Three-phase bolted	99.1	✓Complete	5
07	P1-87L	87L GFL 25% penetration	95.8	✓Complete	5
08	P1-87L	87L GFL 50% penetration	94.3	✓Complete	5
09	P1-87L	87L GFL 75% penetration	92.7	✓Complete	5
10	P1-87L	87L CT saturation compensation	91.4	✓Complete	5
11	P1-87L	87L GFM source infeed	96.1	✓Complete	5
12	P1-87L	87L PV array mixed infeed	95.2	✓Complete	5
13	P1-87L	87L BESS infeed	94.8	✓Complete	5
14	P1-87L	87L Mixed SG+GFL+GFM	93.9	✓Complete	5
15	P1-87L	87L DFIG slip-harmonic	92.4	✓Complete	5
16	P1-87L	87L Pilot channel delay	94.0	✓Complete	5
17	P1-87L	87L EKF parameter convergence	96.5	✓Complete	4
18	P1-87L	87L RLS impedance tracking	95.1	✓Complete	4
19	P1-87L	87L SPRT change detection	93.7	✓Complete	4
20	P1-87L	87L Full SAMBP integration test	94.8	✓Complete	5
21	P2-87T	87T Inrush discrimination	97.3	✓Complete	5
22	P2-87T	87T Internal phase fault	98.1	✓Complete	5
23	P2-87T	87T Cross-country fault	95.4	✓Complete	5
24	P2-87T	87T Over-excitation	94.2	✓Complete	5
25	P2-87T	87T Turn-to-turn fault	90.2	✓Complete	5
26	P2-87T	87T Sympathetic inrush	93.8	✓Complete	5
27	P2-87T	87T IBR-fed transformer	94.7	✓Complete	5
28	P2-87T	87T GFM-fed inrush	96.0	✓Complete	5
29	P2-87T	87T DFIG-fed transformer	93.5	✓Complete	5
30	P2-87T	87T GIC half-cycle saturation	91.8	✓Complete	5
31	P2-87T	87T Restricted earth fault	96.8	✓Complete	5

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Table 13.1 continued

TR#	Phase	Title	Agree%	Status	Ch.
32	P2-87T	87T SPRT inrush/fault	95.5	✓Complete	5
33	P2-87T	87T PINN parameter ID	94.9	✓Complete	4
34	P2-87T	87T EKF core saturation	93.3	✓Complete	4
35	P2-87T	87T Full integration test	95.8	✓Complete	5
36	P3-Dist	Distance Mho SG-only	97.5	✓Complete	6
37	P3-Dist	Distance Mho GFL infeed	91.4	✓Complete	6
38	P3-Dist	Distance Mho GFM infeed	93.2	✓Complete	6
39	P3-Dist	Distance Mho mixed infeed	89.8	✓Complete	6
40	P3-Dist	Distance Quadrilateral SG	96.1	✓Complete	6
41	P3-Dist	Distance Quad GFL	90.7	✓Complete	6
42	P3-Dist	Distance DFIG power swing	88.4	✓Complete	6
43	P3-Dist	Distance load encroachment	91.5	✓Complete	6
44	P3-Dist	Distance series compensation	89.1	✓Complete	6
45	P3-Dist	Distance adaptive IBR reach	93.7	✓Complete	6
46	P3-Dist	Distance GFM-islanded	87.1	✓Complete	6
47	P3-Dist	Distance reverse fault	94.8	✓Complete	6
48	P3-Dist	Distance high-Z fault	90.3	✓Complete	6
49	P3-Dist	Distance zone-2 time-delay	92.0	✓Complete	6
50	P3-Dist	Distance full integration	93.5	✓Complete	6
51	P3-87B	Bus differential SG-only	97.2	✓Complete	5
52	P3-87B	Bus diff IBR infeed	94.5	✓Complete	5
53	P3-87B	Bus diff CT ratio mismatch	91.8	✓Complete	5
54	P3-87B	Bus diff through-fault stability	89.5	✓Complete	5
55	P3-87B	Bus diff full integration	94.1	✓Complete	5
56	P4-Gen	DFIG OC coordination	93.2	✓Complete	7
57	P4-Gen	87T SPRT inrush integration	94.0	✓Complete	5
58	P4-Gen	Pole-slip Relay 78	91.7	✓Complete	7
59	P4-Gen	GFL OC blinding	88.6	✓Complete	7
60	P4-Gen	LOF Relay 40	92.5	✓Complete	7
61	P4-Gen	Ground fault Relay 64G	94.8	✓Complete	7
62	P4-Gen	PV array OC	91.2	✓Complete	7
63	P4-Gen	Frequency Relay 81 ROCOF	93.6	✓Complete	7
64	P4-Gen	87T SPRT phase-phase	95.1	✓Complete	5
65	P4-Gen	Generator protection suite	92.8	✓Complete	7
66	P4-Gen	Anti-islanding GFM	90.4	✓Complete	7
67	P4-Gen	Gen protection integration	93.1	✓Complete	7
68	P5-Mic	BESS microgrid 87L	95.8	✓Complete	8
69	P5-Mic	Islanded microgrid OC	91.0	✓Complete	8
70	P5-Mic	Microgrid 87B	93.4	✓Complete	8
71	P5-DT	DT shadow mode validation	96.2	✓Complete	9
72	P5-DT	DT predict mode EKF	95.7	✓Complete	9
73	P5-DT	DT scenario library KNN	94.3	✓Complete	9
74	P5-DT	DC microgrid di/dt protection	92.1	✓Complete	8
75	P5-DT	DT Bayesian fusion	95.1	✓Complete	9
76	P5-DT	DT 87L full validation	96.2	✓Complete	9
77	P5-DT	DT adaptation engine	94.8	✓Complete	9
78	P5-Spc	SSCI detector DFIG	90.7	✓Complete	8
79	P5-Spc	Sub-synchronous OC	89.5	✓Complete	8
80	P5-Spc	DC microgrid mirror	93.0	✓Complete	8
81	P5-Spc	GMD/GIC LP solver	91.4	✓Complete	8
82	P5-Spc	GOOSE anomaly IEC62351	97.8	✓Complete	10
83	P5-Spc	Adaptive OC CLPU pre-study	92.5	✓Complete	8
84	P5-Spc	Ferroresonance 20-scenario	88.4	✓Complete	8

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Table 13.1 continued

TR#	Phase	Title	Agree%	Status	Ch.
85	P5-Spc	CLPU adaptive OC coordina- tor	91.8	✓Complete	8

Chapter 14

Future Research: TR-86 to TR-100

14.1 Roadmap Overview

Table 14.1: Planned TRs 86–100

TR	Target Date	Title
TR-86	Q3 2026	Multi-terminal line differential (87L-MT)
TR-87	Q3 2026	HVDC protection — pole-to-ground
TR-88	Q4 2026	Offshore wind farm aggregated model
TR-89	Q4 2026	Virtual impedance GFM LOF
TR-90	Q1 2027	Federated learning for relay coordination
TR-91	Q1 2027	Quantum-safe GOOSE authentication
TR-92	Q2 2027	Reinforcement learning adaptive thresholds
TR-93	Q2 2027	Wide-area back-up protection
TR-94	Q3 2027	6th-order GFM VSM protection
TR-95	Q3 2027	Hydrogen electrolyser fault current
TR-96	Q4 2027	STATCOM protection
TR-97	Q4 2027	Multi-infeed HVDC interaction
TR-98	Q1 2028	DT hardware-in-the-loop (HIL)
TR-99	Q2 2028	Regulatory framework (IEC 60255-187)
TR-100	Q3 2028	Thesis final validation & submission

14.2 TR-86: Multi-Terminal 87L

Multi-terminal line differential requires generalising the two-terminal trip logic (Eq. 5.1) to N terminals:

$$I_{diff}^{MT} = \left| \sum_{k=1}^N \mathbf{I}_k \right|, \quad I_{rest}^{MT} = \sum_{k=1}^N |\mathbf{I}_k| \quad (14.1)$$

The challenge is synchronising N phasor measurements with latency τ_k each. SAMBP will extend the EKF to estimate communication delay as an additional state variable.

14.3 TR-90: Federated Learning

Relays in different substations can share gradient updates without revealing raw fault data:

$$\mathbf{w}^{(t+1)} = \sum_{k=1}^K \frac{n_k}{N} \mathbf{w}_k^{(t)} \quad (14.2)$$

where $\mathbf{w}_k^{(t)}$ is relay k 's local model weight, n_k is local data size.

14.4 TR-92: Reinforcement Learning

Formulate threshold adaptation as a Markov Decision Process:

- **State:** $\mathbf{s}_t = [\rho_{IBR}, R_F, P_{load}, \text{agree_rate}_{t-1}]$
- **Action:** $\mathbf{a}_t = [k_r, I_{min}, k_1, k_2]$ (relay parameters)
- **Reward:** $r_t = \text{agree_rate}_t - \lambda_{maltrip} \cdot \mathbb{1}[\text{mal-trip}]$

Appendix A

Key Equations Reference

Table A.1: Master equation reference

Symbol / Equation	Description	Eq. No.
$i_a(t) = \dots$	SG subtransient fault current	3.1
$\mathbf{x}_{SG}$	SG Park-dq state vector	—
I_{fault}^{GFL}	GFL current-limited fault current	3.8
$i_s(t)$	DFIG three-component fault current	3.13
I_{fault}^{BESS}	BESS fault current	3.15
EKF predict/update	Eqs. 4.3–4.7	—
$\mathcal{L}_{PINN}$	PINN combined loss	4.11
Λ_k	SPRT likelihood ratio	4.12
I_{diff}, I_{rest}	87L differential/restraint	5.1–5.2
Mho criterion	Distance zone geometry	6.3
Z_{app}	Apparent impedance	6.1
I_{GIC}	Geomagnetic induced current	5.10
$i_{dc}(t)$	DC fault transient	8.1
f_{SSCI}	Sub-synchronous resonance frequency	8.3
Λ	Lyapunov exponent (ferroresonance)	8.4
I_{CLPU}	Cold load pickup current	8.5

Appendix B

Algorithm Pseudocode Index

Table B.1: Algorithm index

Algorithm	Name	Page
1	EKF Parameter Estimator	12
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Appendix C

Symbol Glossary

Table C.1: Symbol glossary

Symbol	Definition
X_d'', X_d', X_d	Subtransient, transient, steady-state d-axis reactances
$\tau_d'', \tau_d', \tau_a$	Corresponding time constants
E_q, E_d	Internal voltage components
δ	Rotor angle
ω_0	Nominal angular frequency ($2\pi \times 50$ rad/s)
I_{max}	IBR maximum current limit (p.u.)
τ_{cc}	Current-control loop time constant
s	DFIG slip
R_F	Fault resistance
ρ_{IBR}	IBR penetration ratio
P, Q, R	EKF covariance matrices
K	Kalman gain matrix
λ	RLS forgetting factor
λ_p, λ_r	PINN penalty weights
Λ	SPRT log-likelihood ratio
α, β	SPRT false alarm / miss probability
k_r	87L slope factor
I_{min}	87L minimum pickup current
T_M	SAMBP protection time window (50 ms)
H	Inertia constant (s)
$ROCOF$	Rate of change of frequency (Hz/s)
I_{GIC}	Geomagnetic induced current
$\theta_{hotspot}$	Transformer hotspot temperature
f_{SSCI}	Sub-synchronous resonance frequency
$\Lambda_{Lyapunov}$	Lyapunov exponent
k_{cold}	Cold load pickup factor
$stNum, sqNum$	GOOSE state/sequence numbers

Appendix D

Key File Paths

Table D.1: SAMBP key file paths

Component	Path
Project root	/root/phd.thesis/04_code/sambp/
87L module	sambp/line_diff/
87T module	sambp/transformer_diff/
87B module	sambp/bus_diff/
Distance	sambp/distance/
Generator (40)	sambp/generator/relay40/
Generator (64G)	sambp/generator/relay64g/
Generator (78)	sambp/relay78/
Digital Twin	sambp/digital_twin/
DT Reports	sambp/digital_twin/reports/figures/
TR68 Figures	.../figures/TR68/
TR76 Figures	.../figures/TR76/
TR86+ Stubs	sambp/TR86/
Sync OC	sambp/sync_oc/
Coordination	sambp/coordination/
System	sambp/system/

Appendix E

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